Tutoraggio: Venerdì 24/5 ore 14.30 – 17.30 Aula Ex D

~~Lunedì 27/5~~

IPOTESI LAVORO

$$\gamma: a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + 2a_{13}x + 2a_{23}y + a_{33} = 0$$

γ NON degenere (cioè, $\det(A') \neq 0$)

$\boxed{\text{Caso } a_{12} = 0}$ Basta una traslazione...

metodo complet. quadrati

$$\to \gamma: \underbrace{c_1(x-x_0)^2}_{\uparrow} + \underbrace{c_2(y-y_0)^2}_{\uparrow} + \text{termini grado inferiore}$$

Traslazione $\begin{cases} x' = x - x_0 \\ y' = y - y_0 \end{cases}$

$$\gamma: 4x^2 - y^2 + 8x + 2y - 13 = 0$$

$$\gamma: 4(x^2 + 2x + 1) - 4 - (y^2 - 2y + 1) + 1 - 13 = 0$$

$$\gamma: 4(x+1)^2 - (y-1)^2 = 16 \qquad \begin{cases} x' = x + 1 \\ y' = y - 1 \end{cases}$$

$$\gamma: \frac{x'^2}{4} - \frac{y'^2}{16} = 1$$

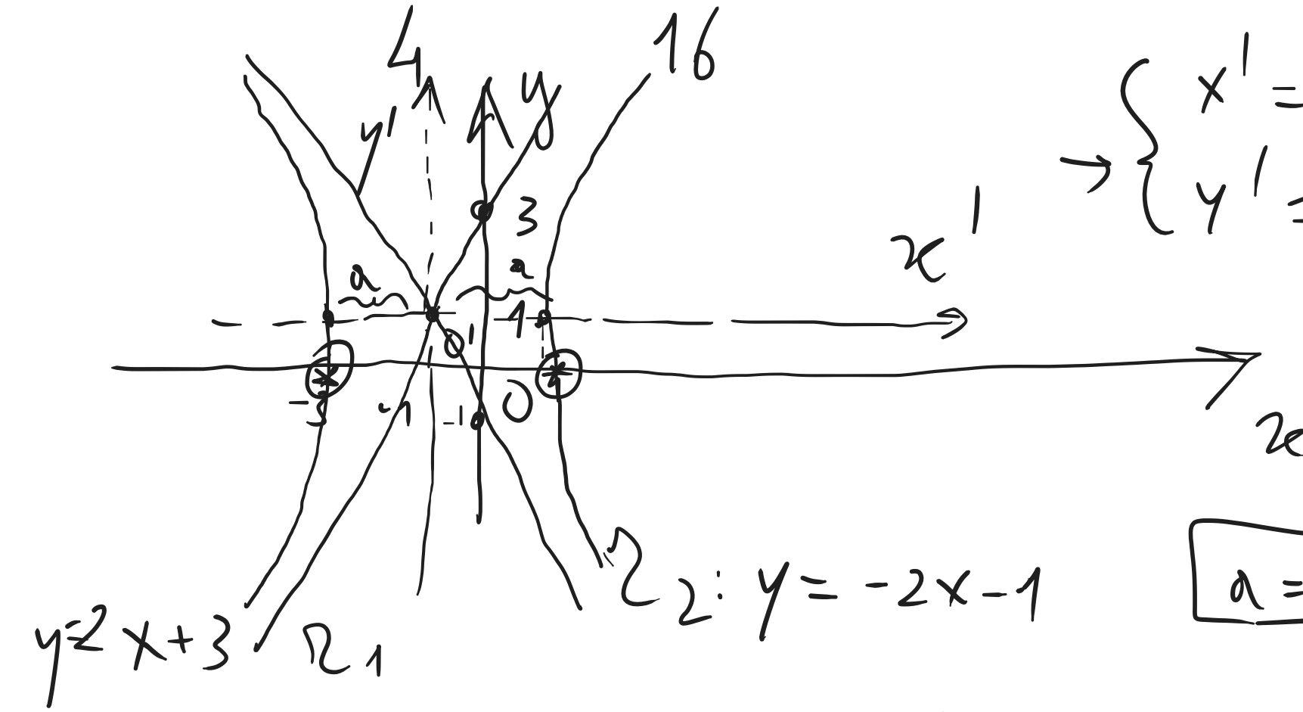
iperbole con $a = 2$ $b = 4$ $\frac{b}{a} = 2$

$O' = [-1, 1]$ rispetto al sistema Oxy

$\to \begin{cases} x' = x + 1 \\ y' = y - 1 \end{cases}$

Asintoti: $\boxed{r_1: y' = 2x'}$ $\boxed{y' = \pm\frac{b}{a}x'}$

$r_2: y' = -2x'$



$\boxed{a = 2}$

Asintoti rispetto a x, y: $r_1: y - 1 = 2(x+1)$ cioè $y = 2x + 3$

$r_2: y - 1 = -2(x+1)$ " $y = -2x - 1$

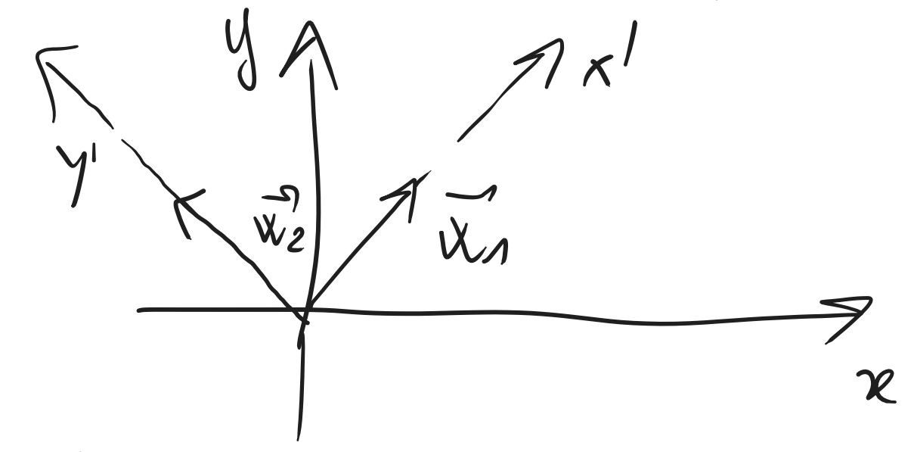
$\circledast \begin{cases} \gamma \\ y = 0 \end{cases}$

$\boxed{\text{Caso } a_{12} \neq 0}$ (Rotazione + eventuale traslazione)

$$\gamma: \underbrace{a_{11}x^2 + 2a_{12}xy + a_{22}y^2}_{} + 2a_{13}x + 2a_{23}y + a_{33} = 0$$

$$\sqcup = [x, y]\, A \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{dove } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix} \quad A = {}^tA$$

$\exists$ rotazione $P = [\vec{w}_1, \vec{w}_2]$ t.c. ${}^tP.A P = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$



$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix} \quad [x, y] = [x', y']\, {}^tP$$

Rispetto alle coord. ruotate x', y' la parte quadratica $\sqcup$ diventa:

$$\sqcup = [x, y]\, A \begin{bmatrix} x \\ y \end{bmatrix} = [x', y']\, {}^tP A P \begin{bmatrix} x \\ y \end{bmatrix} = [x', y'] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$= \lambda_1 x'^2 + \lambda_2 y'^2$$

Conclusione: nel sistema ruotato $Ox'y'$ siamo rientrati nel caso $a_{12} = 0$ $\gamma: \lambda_1 x'^2 + \lambda_2 y'^2 + \text{termini grado inf} = 0$

Es: $\gamma: \underbrace{7x^2 - \overbrace{6\sqrt{3}xy}^{a_{12} \neq 0} + 13y^2}_{} - 16 = 0$ (γ NON deg.) $\det A' \neq 0$

$$\sqcup = [x, y]\, A \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{dove } A = \begin{bmatrix} 7 & -3\sqrt{3} \\ -3\sqrt{3} & 13 \end{bmatrix} \quad \boxed{-6\sqrt{3} = 2a_{12}}$$

Dobbiamo determinare rotazione P t.c. ${}^tPAP = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

$$\underbrace{p(\lambda)}_{} = |A - \lambda I| = \begin{vmatrix} (7-\lambda) & -3\sqrt{3} \\ -3\sqrt{3} & (13-\lambda) \end{vmatrix} = \lambda^2 - 20\lambda + 91 - 27 = $$

$$= \underbrace{\lambda^2 - 20\lambda + 64}_{}$$

$\text{vers}(\vec{u}) = \frac{\vec{u}}{|\vec{u}|}$

$\lambda_1 = 4$ $\lambda_2 = 16$ (ellisse)

Dobbiamo costruire $P = [\vec{w}_1, \vec{w}_2]$ t.c. ${}^tP.A.P = \begin{bmatrix} 4 & 0 \\ 0 & 16 \end{bmatrix}$ $\underbrace{A - \lambda_1 I}_{\rho = 1} = \begin{bmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & 9 \end{bmatrix}$

$\vec{w}_1$ si ricava da $(A - \lambda_1 I)\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, cioè

$\begin{cases} 3x - 3\sqrt{3}y = 0 \\ \cancel{-3\sqrt{3}x + 9y = 0} \Leftarrow \text{perché } \rho = 1 \end{cases}$ $\vec{w}_1 = \text{vers}\left(\begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}\right) = \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$

$\vec{w}_2$ si ricava da $(A - \lambda_2 I)\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \xrightarrow{\text{verificare}} \vec{w}_2 = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$

$\vec{w}_1 = \begin{bmatrix} a \\ b \end{bmatrix}$ $\vec{w}_2 = \begin{bmatrix} -b \\ a \end{bmatrix}$ Conclusione $P = [\vec{w}_1, \vec{w}_2] = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \frac{\sqrt{3}}{2} \end{bmatrix}$

$\det P = 1$ o.k.

Rispetto alle coordinate ruotate x', y' legate alle coord. x, y da

$\to \circledast \begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix}$ l'eq. di γ diventa:

1° grado $\downarrow$ $\sqcup = \ldots = \lambda_1 x'^2 + \lambda_2 y'^2$

$$\gamma: \underbrace{7x^2 - 6\sqrt{3}xy + 13y^2}_{} - 16 = 0 \qquad \circledast \begin{cases} x = \frac{\sqrt{3}}{2}x' - \frac{1}{2}y' \\ y = \frac{1}{2}x' + \frac{\sqrt{3}}{2}y' \end{cases}$$

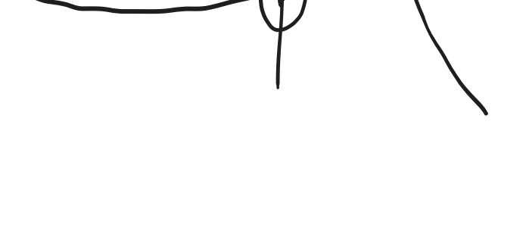
1° grado (NON qui)

$$\gamma: \underbrace{4x'^2 + 16y'^2}_{} - 16 = 0$$

$$\gamma: \frac{x'^2}{a^2} + \frac{y'^2}{b^2} = 1$$

ellisse $a = 2$ $b = 1$

$\vec{w}_1 = \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$



$\odot \begin{cases} \gamma \\ x = 0 \end{cases}$ $\vec{w}_2 = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$

$\circledast \begin{cases} \gamma \\ y = 0 \end{cases}$